

```
In [2]: def bisection(f, a, b, tol):
        if f(a) * f(b) >= 0:
            print("The function must have opposite signs at a and b.")

        while (b - a) / 2 > tol:
            c = (a + b) / 2
            if f(c) == 0:
                return c

            if f(a) * f(c) < 0:
                b = c
            else:
                a = c

        return (a + b) / 2
```